

# Technical Document #382: Software Engineering - Comprehensive Overview

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API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Design patterns provide reusable solutions to common software design problems. Singleton, factory, and observer patterns are widely used.

Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

### Key Takeaways:

- Code review examines code changes before merging.
- API design defines how software components communicate.
- Refactoring improves code structure without changing behavior.